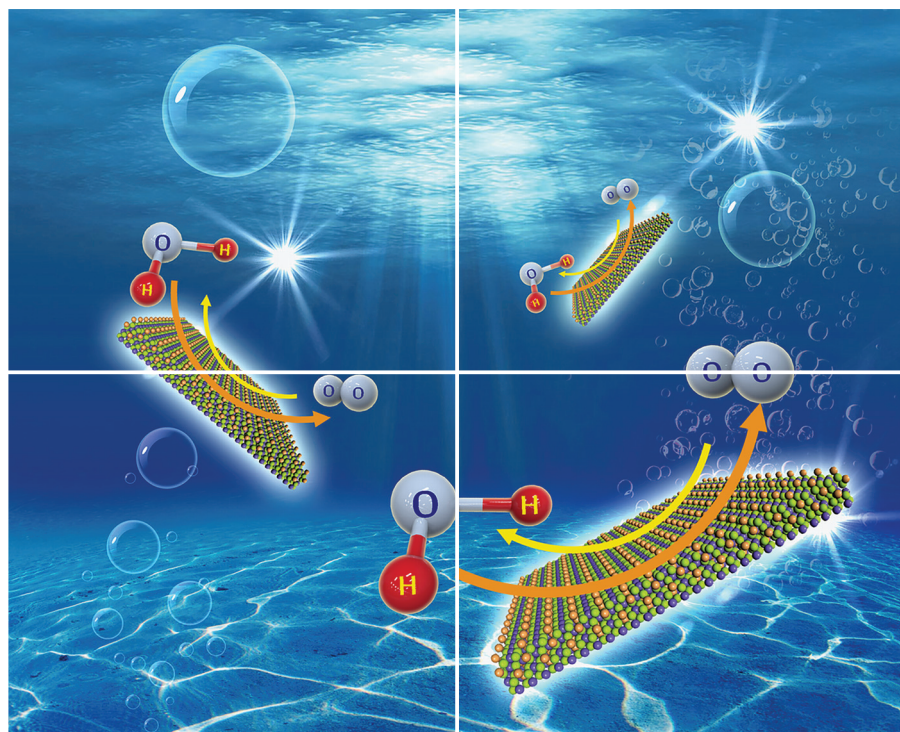




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Enhanced thermoelectric properties of the Dirac semimetal Cd_3As_2 [†]

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We report an investigation of temperature- and doping-dependent thermoelectric behavior of the topological semimetal Cd_3As_2 . The electrical conductivity, thermal conductivity, Seebeck coefficient, and figure of merit (ZT) are calculated using the Boltzmann transport theory. The calculated thermoelectric properties of the pristine Cd_3As_2 match well with the experimental results. Electron or hole doping, especially the latter, is found to much improve the thermoelectric behavior of the material. The optimum figure of merit ZT of Cd_3As_2 with electron doping is found to be about 0.5 at $T = 700$ K with $n = 1 \times 10^{20} \text{ cm}^{-3}$, which is much larger than the maximum experimental value obtained for pristine Cd_3As_2 (~ 0.15). For p-type Cd_3As_2 , the maximal value of the Seebeck coefficient as a function of temperature increases apparently with the increase of the hole doping concentration and its position shifts drastically towards the lower temperature region, compared to that of n-type Cd_3As_2 . This leads to an optimum figure of merit ZT of about 0.5, obtained at a low temperature of 500 K ($p = 1 \times 10^{20} \text{ cm}^{-3}$) in the p-type Cd_3As_2 .

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1. Introduction

Materials exhibiting thermoelectric performance are of considerable importance in industry due to their numerous potential applications, including in vehicular exhaust waste-heat recovery, energy harvesting, heating and cooling, and solid-state energy conversion.^{1,2} High thermoelectric performance is extremely desirable for these applications.^{3,4} Thermoelectric behavior is typically quantified in terms of a dimensionless figure of merit (ZT) given by the expression $ZT = \frac{S^2\sigma T}{\kappa}$, in which S is the Seebeck coefficient or thermopower, σ is the electrical conductivity, T is the absolute temperature, and κ is the thermal conductivity, normally the sum of the electronic and lattice contributions, namely $\kappa = \kappa_e + \kappa_L$. $S^2\sigma$ is the power factor of the thermoelectric performance. The ZT expression shows that for good thermoelectric performance both the Seebeck coefficient and electrical conductivity are expected to

be high, while the thermal conductivity is expected to be low. These parameters are, however, intimately coupled to each other, resulting in conflicts in the optimization process.^{1,2} Hence, optimizing and finding usable high-performance thermoelectric materials remains full of challenges.

Recently, three-dimensional (3D) Dirac semimetal states have been theoretically predicted and experimentally realized in Cd_3As_2 .^{5–9} Unlike other semimetals, the Cd_3As_2 crystal possesses Dirac fermions that disperse linearly in k -space and as a result it becomes a crystalline material with an ultrahigh electron mobility, μ , of about 10^4 – $10^6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.^{8,9} Since the power factor strongly depends on the electron mobility *i.e.*, $S^2\sigma \approx \mu(m^*/m_e)^{1.5}$ (where m^* is the energy-band electron effective mass and m_e is the free electron mass),¹⁰ Cd_3As_2 shows great potential for high performance thermoelectric applications. The maximum ZT value of pristine Cd_3As_2 that has been achieved in experiments is, however, only about 0.15.^{11,12} How to improve the thermoelectric performance in this material is of significance. Semimetals usually can be easily doped with both electrons and holes due to the existence of Dirac cones near the Fermi level (E_F), which has been proved in graphene.¹³ Both the n-type^{11,12,14,15} and p-type^{16,17} Cd_3As_2 have also been successfully synthesized in experiments. It is meaningful to explore systematically the effect of the electron or hole doping on the thermoelectric properties of the semimetal Cd_3As_2 .

In this work, we perform first principles computations on the electronic states and thermoelectric properties of the topological semimetal Cd_3As_2 . It is found that the electronic structure of Cd_3As_2 can be described correctly by a generalized

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[†]Electronic supplementary information (ESI) available: Calculated electrical conductivity (σ), Seebeck coefficient (S), thermal conductivity (κ), and figure of merit (ZT) of n/p-type Cd_3As_2 with respect to the chemical potential at different temperatures and calculated densities of states for the Cd_3As_2 crystal. See DOI: 10.1039/C6QI00383D

gradient approximation after the spin-orbit coupling is considered. Based on the Boltzmann transport theory, we calculated the electrical conductivity, thermal conductivity, Seebeck coefficient, and figure of merit of the semimetal. The calculated thermoelectric properties of the pristine Cd_3As_2 match well with the experimental results. Very interestingly, the ZT value of the semimetal is found to be much improved by both electron and hole doping, especially the latter. For the n-type Cd_3As_2 , the optimal doping level is about $1 \times 10^{20} \text{ cm}^{-3}$, leading to the ZT increasing to about 0.5 at $T = 700 \text{ K}$, which is much larger than the ZT (0.15) of the pristine Cd_3As_2 . In contrast, for the p-type Cd_3As_2 , the maximal value of the Seebeck coefficient shifts towards the low temperature region and the S value in the hole-doping region is much larger than the one in the electron-doping region, resulting in an optimum ZT of about 0.5 at 500 K for the p-type Cd_3As_2 . Our work provides new paths towards high thermoelectric performance in topological Dirac semimetals.

II. Methods and models

The calculations were performed using the general potential linearized augmented plane-wave (LAPW) method¹⁸ as implemented in the WIEN2K code.¹⁹ Spin-orbital coupling is included as a second vibrational step using scalar-relativistic eigenfunctions as the basis after the initial calculation is converged to self-consistency. The convergence of the calculations regarding the size of the basis set is achieved using an $R_{\text{MT}}K_{\text{max}}$ value of 7, where R_{MT} is the smallest atomic sphere radius in the unit cell and K_{max} is the magnitude of the largest

K wave vector inside the first Brillouin zone. Exchange and correlation effects are accounted for using the generalized gradient approximation Perdew–Burke–Ernzerhof (PBE) functional.²⁰ Considering the possible underestimation of the band gap by PBE, we do the calculation based on the modified Becke–Johnson (mBJ) potential,²¹ which has been proved to be a good correction to the exchange and correlation functional for electronic structure calculations of thermoelectric systems.^{22–24} From the calculated band structure, the thermoelectric transport tensors are evaluated using the semi-classical Boltzmann kinetic transport theory within the constant relaxation time approximation (CSTA) and the rigid band approach as implemented within the BoltzTraP program.²⁵ The constant relaxation time is given as a standard electron–phonon model,^{22,26} the parameter of which is given by fitting the experimental data. This method has been successfully used to describe the transport coefficients of a wide range of thermoelectric materials.^{27–32}

III. Results and discussion

Cd_3As_2 has a distorted superstructure of the antifluorite (M_2X) structure type with an $I4_1acd$ space group.⁵ It can be viewed as a tetragonally-distorted anti-fluorite structure with 1/4 M sites of vacancies and contains 80 atoms per unit cell. The calculated electronic band structures without and with spin-orbit coupling (SOC) based on the PBE potentials are shown in Fig. 1(a) and (b), respectively. As shown in Fig. 1(a), the conduction and valence bands of Cd_3As_2 are degenerate at the Γ point without the SOC. Similar to most semiconductors with

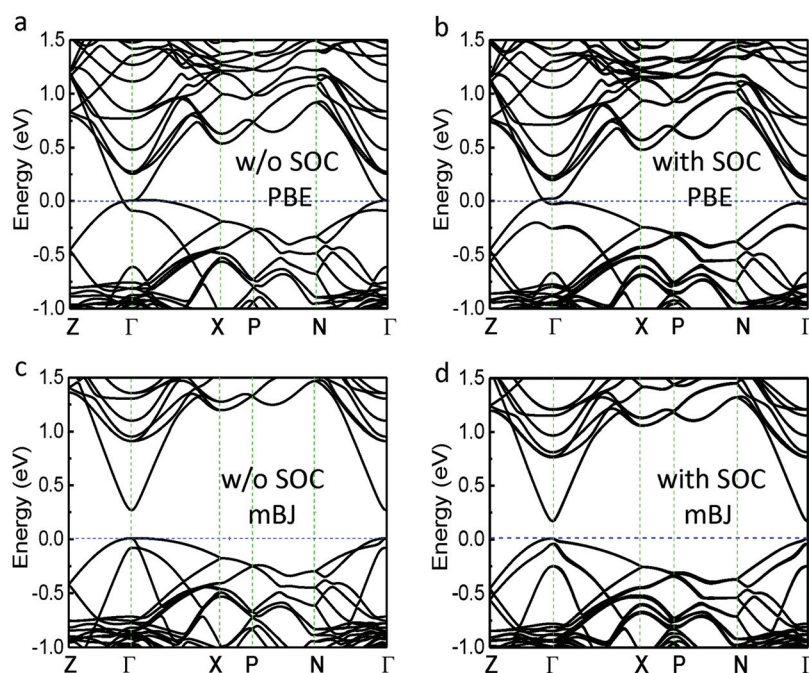


Fig. 1 (a) and (b) Calculated band structures for Cd_3As_2 using PBE without and with SOC being considered, respectively. (c) and (d) Calculated band structures for Cd_3As_2 using mBJ without and with SOC being considered, respectively.

antifluorite or zinc-blende structures, the low energy electronic properties of Cd_3As_2 are mostly determined by the Cd 5s states (conduction bands) and the As 4p states (valence bands). When the SOC is considered, the bands are inverted around the Γ point with the s states lower than the p states, which is an important sign of the nontrivial topology appearance. In contrast to ordinary topological insulators, no global band gap opens in the Cd_3As_2 even after the SOC is taken into account.^{5,8} Instead the bands cross around the Γ point along the Γ -Z direction, exactly at the E_F , due to the protection of the C_{4v} symmetry with respect to the k_z axis. This unique band characteristic, together with the time-reversal and space inversion symmetries, gives rise to a 3D massless Dirac semimetal property of the system, in good agreement with the results of ref. 5 and 8. Since a material's thermoelectric properties strongly depend on the band structure,²⁵ and the mBJ potential was proved to be an excellent method for calculating the electric states of the thermoelectric systems,^{22–24} we did the calculation for Cd_3As_2 based on the mBJ potential. The electronic band structures without and with SOC based on the mBJ potential are given in Fig. 1(c) and (d), respectively. A global gap of 160 meV is opened at the Γ point in Fig. 1(d), which is inconsistent with the experimental data,^{8,9} indicating that the mBJ potential cannot describe well the electronic structure of 3D Dirac semimetals and may even produce a wrong result. This observation might be explained by the fact that the mBJ potential was designed to reproduce the shape of the exact exchange optimized effective potential of atoms.^{21,22,24} Because the mBJ potential leads to an increase of the gap values, the conduction bands and valence bands are separated too much, making the Dirac point disappear. It can

be expected that if the gap opened by the mBJ in Fig. 1(c) is not very big and the band inversion can still be induced by the SOC interaction, the 3D Dirac semimetal behavior will remain in the system. The reason for this is that the mBJ potential does not break the C_{4v} symmetry in the system.

Since the electronic structure of Cd_3As_2 is described correctly with the PBE functional, the thermoelectric properties of Cd_3As_2 were investigated using the Boltzmann transport theory based on the band structures obtained with the PBE functional (Fig. 1(b)). In experiments, the fabricated intrinsic Cd_3As_2 is an n-type material with an electron carrier concentration of about $n = 1 \times 10^{19} \text{ cm}^{-3}$,^{11,12} also called pristine Cd_3As_2 in this work. Thus, we first calculated the thermoelectric properties of Cd_3As_2 at this electron carrier concentration. Within the framework of the Boltzmann transport theory, the scattering time relaxation (τ) is usually adopted approximately as a constant, which is associated with the behavior of the electrical conductivity (σ), thermal conductivity from electronic contributions (k_e), and furthermore the ZT of the system.²⁵ The relaxation time generally depends on both the charge carrier concentration (n) and the temperature (T). We here employed a standard electron-phonon dependence on T and n for τ , namely, $\tau = \frac{C}{Tn^{1/3}}$,^{22,26} where C is a constant and can be determined by comparing it to experimental data. For this pristine Cd_3As_2 sample (with $n = 1 \times 10^{19} \text{ cm}^{-3}$), the experimental electrical conductivity is about 67 S cm^{-1} at 300 K,^{11,12} which together with the ratio of σ/τ which was obtained from the Boltzmann transport theory gives a value of C of about $5 \times 10^{-6} \text{ s K cm}$.

The calculated electrical resistivity of the pristine Cd_3As_2 sample with respect to the temperature is plotted in Fig. 2(a).

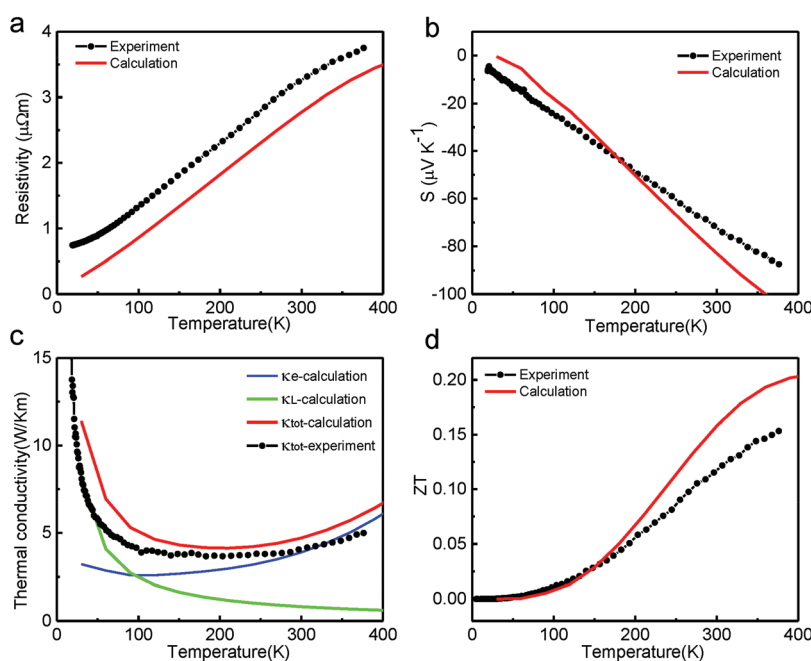


Fig. 2 (a)–(d) Calculated temperature-dependent resistivity, Seebeck coefficient (S), thermal conductivity, and figure of merit (ZT) of Cd_3As_2 , compared with the experimental data from ref. 11.

It shows metallic behavior at low temperatures, which is in good agreement with the experiments, indicating that the standard electron–phonon model is suitable to describe the mechanism of electron–phonon scattering in Cd_3As_2 . Note that since the experimental data of Cd_3As_2 ranged from 0 K to 380 K, we plotted the thermoelectric properties obtained in our calculations also in this temperature range, as shown in Fig. 2, to compare with the experimental results directly. With the constant relaxation time approximation, the Seebeck coefficient is independent of τ . The obtained Seebeck coefficient of the Cd_3As_2 sample also matches the experimental data very well (Fig. 2(b)), meaning that the semiclassical Boltzmann transport theory can be adopted to describe the thermoelectric transport properties of Cd_3As_2 . The calculated electron thermal conductivity (κ_e), plotted in Fig. 2(c), increases with temperature. To evaluate the ZT of this Cd_3As_2 sample, the lattice thermal conductivity (κ_L) is also required, which is obtained by subtracting κ_e from the total thermal conductivity (κ_{tot}) provided from experiments.¹¹ We find that the obtained κ_L follows a classic A/T dependence with $A = 245 \text{ W m}^{-1}$, as shown in Fig. 2(c). At high temperatures, κ_L decreases, giving rise to κ_e contributing more to the total thermal conductivity. Based on the obtained σ , S , and κ_{tot} , the thermoelectric figure of merit can be estimated through the formula $ZT = S^2\sigma T/\kappa_{\text{tot}}$ (Fig. 2(d)). The trend of the calculated ZT , especially in the low temperature region, is in good agreement with the experimental results.^{11,12} The maximum of the calculated ZT is about 0.2 at 400 K, which is also very close to the experimental values.^{11,12} These good agreements indicate that the Boltzmann transport theory with a constant relaxation time

approximation can describe very well the thermoelectric transport properties of Dirac semimetals.

To investigate the thermoelectric properties of Cd_3As_2 with different electron carrier concentrations, the rigid-band approach was employed. Fig. 3 shows the evolution of the electrical conductivity with respect to the temperature for various electron carrier concentrations of interest. At a fixed temperature, the electrical conductivity increases drastically with the carrier concentration. When the concentration is $n = 1 \times 10^{21} \text{ cm}^{-3}$, the electrical conductivity increases to $3.81/\mu\Omega \text{ m}$ at 300 K, which is much higher than that of the pristine Cd_3As_2 ($2.62/\mu\Omega \text{ m}$ at 300 K). The calculated Seebeck coefficient S (Fig. 3(b)) has, however, different trends as a function of the temperature for various electron carrier concentrations. When the electron concentration is low ($n \leq 5 \times 10^{19} \text{ cm}^{-3}$), the absolute value of the calculated S ($|S|$) increases with the temperature and then decreases. The maximal value of $|S|$ increases with concentration and its temperature position shifts up. For high electron concentrations ($n > 1 \times 10^{20} \text{ cm}^{-3}$), the calculated $|S|$ increases almost linearly with temperature in the range below 900 K, like a metal.¹ As a consequence, for the n-type doping, the maximum of the calculated $|S|$ value is about $170 \mu\text{V K}^{-1}$ with $n = 5 \times 10^{19} \text{ cm}^{-3}$ at 700 K. Combining the electron thermal conductivity and lattice thermal conductivity, we plotted the total thermal conductivity with respect to the temperature at various concentrations in Fig. 3(c). Since the κ_L value decreases drastically with temperature ($\kappa_L = A/T$), the trend of κ_{tot} is determined by the κ_e for temperatures larger than 200 K. With the obtained σ , S , and κ_{tot} , the figure of merit of the material at various electron concentrations can be

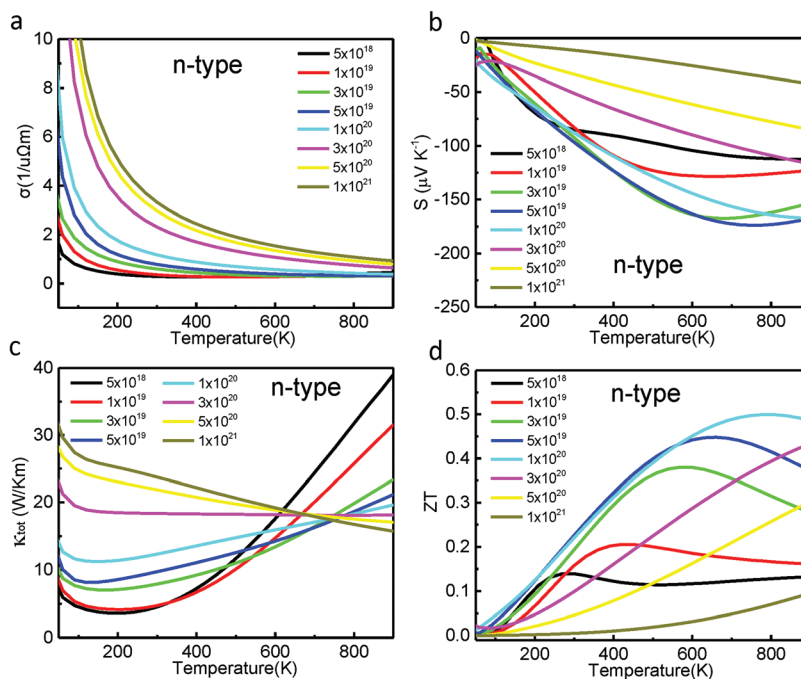


Fig. 3 (a)–(d) Calculated electrical conductivity (σ), Seebeck coefficient (S), thermal conductivity (κ), and figure of merit (ZT) of n-type Cd_3As_2 with respect to the temperature for different carrier concentrations (cm^{-3}).

calculated, which is presented in Fig. 3(d). Obviously, the ZT shares a similar tendency to $|S|$ as a function of concentration (Fig. 3(b)) to a certain extent. The maximum optimum ZT of Cd_3As_2 is found to be about 0.5 at $T = 700$ K with $n = 1 \times 10^{20} \text{ cm}^{-3}$. This carrier concentration ($n = 1 \times 10^{20} \text{ cm}^{-3}$) is hopefully achieved in experiments with current advanced technologies.^{11,16,33} This predicted ZT value (0.5) of Cd_3As_2 with electron doping is much larger than the maximum ZT (0.15) achieved in the experiments for the pristine samples.^{11,12}

The thermoelectric behavior of Cd_3As_2 with hole doping is also explored since electron and hole doping are both easily carried out in semimetals in experiments.^{16,17} As shown in Fig. 4(a) and (c), the trends of both electronic conductivity and thermal conductivity of the p-type Cd_3As_2 as functions of temperature and concentration are similar to those of the n-type Cd_3As_2 . The magnitudes of σ and κ_{tot} in the p-type Cd_3As_2 are, however, usually lower than those of the n-type Cd_3As_2 at a fixed temperature or concentration, which is associated with the band structures. The S curves in the p-type Cd_3As_2 (Fig. 4(b)) are very different from those of the n-type Cd_3As_2 (Fig. 3(b)). Fig. 4(b) shows that the maximal values of S at various hole concentrations are located at lower temperatures than those of the n-type Cd_3As_2 and the S absolute values in the hole-doping region are much larger than the corresponding ones in the electron-doping region. As a result, the ZT of the p-type Cd_3As_2 is enhanced greatly for the middle temperature region (Fig. 4(d)), compared to that of the n-type sample. For the p-type Cd_3As_2 , an optimum ZT of about 0.5 can be acquired at 500 K with a hole doping concentration of $p = 1 \times 10^{20} \text{ cm}^{-3}$, as shown in Fig. 4(d).

Because the concentrations of Cd_3As_2 are usually tuned by the gate voltage,^{16,17} it is significant to report the thermoelectric properties with respect to chemical potential at various temperatures. Thus, we plot the σ , S , κ_{tot} , and ZT in terms of chemical potential at various temperatures as Fig. S1(a)–(d), respectively, in the ESI.† At the same temperature, the maximum value of σ with electron doping is larger than that with hole doping and the σ of both n-type and p-type Cd_3As_2 decreases with temperature (Fig. S1(a)†), which is consistent with the trends in Fig. 3(a) and 4(a). For the Seebeck coefficient, it is interesting to find that its maximum values of hole doping at different temperatures all occur at about 0.1 eV below the E_F , where the band dispersion is weak (Fig. 1(b)). Meanwhile for the electron doping, the S maximum values all occur at about 0.25 eV above the E_F due to the degenerated bands around this energy (Fig. 1(b)). For most temperatures, the S maximum values of p-type Cd_3As_2 are larger than those of n-type Cd_3As_2 , resulting in a larger ZT in p-type Cd_3As_2 , which is also in agreement with the trends obtained from Fig. 3(d) and 4(d). The reason why the thermoelectric behavior of the p-type Cd_3As_2 is superior to that of the n-type Cd_3As_2 can be ascribed to the band dispersions around the E_F . The relation between temperature and Seebeck coefficient can be seen using relatively simple models of electron transport. For metals or degenerate semiconductors, the Seebeck coefficient is approximately given by:¹

$$S = \frac{8\pi^2 k_B^2 T}{3qh^2} m^* \left(\frac{\pi}{3n} \right)^{2/3},$$

where q is the carrier charge, n is the carrier concentration, and m^* is the effective mass of the carrier. As shown in Fig. 1(b), the

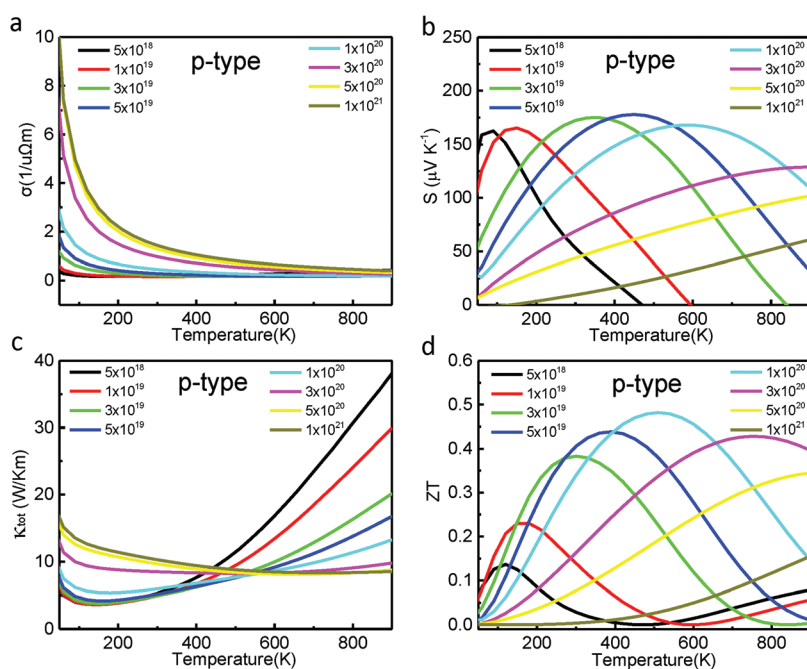


Fig. 4 (a)–(d) Calculated electrical conductivity (σ), Seebeck coefficient (S), thermal conductivity (κ), and figure of merit (ZT) of p-type Cd_3As_2 with respect to the temperature for different carrier concentrations (cm^{-3}).

bands around the valence band maximum (VBM) disperse less than those around the conduction band minimum (CBM) do. A weak dispersion band gives electrons with heavy effective masses, which can enhance the Seebeck coefficient.¹ This trend can also be obtained by the analysis of the densities of states (DOSs) of Cd₃As₂. As shown in Fig. S2 in the ESI,[†] the slope of the total DOS at the VBM is larger than that at the CBM, indicating that the effective mass at the VBM is larger than that at the CBM, which is in agreement with the band analysis. Since the m^* of the p-type is larger than that of the n-type near the E_F , the maximal values of S at various hole concentrations are located at lower temperatures than those of the n-type Cd₃As₂. This trend can also be found in other thermoelectric materials, such as PbTe³⁴ and CuCoO₂.³⁵

IV. Conclusions

In summary, we performed first principles computations on the electronic structure and thermoelectric behavior of the topological semimetal Cd₃As₂ and found that the electronic structure of Cd₃As₂ can be described correctly by the generalized gradient approximation with spin-orbit coupling. Based on the Boltzmann transport theory, the electrical conductivity, thermal conductivity, Seebeck coefficient, and figure of merit of the system were calculated. The optimum ZT of Cd₃As₂ with electron-doping was found to be about 0.5 at $T = 700$ K with $n = 1 \times 10^{20} \text{ cm}^{-3}$, while for the p-type Cd₃As₂, the maximal value of the Seebeck coefficient as a function of temperature increases with the increase of the hole doping concentration and its position shifts drastically towards the lower temperature region compared to that of the n-type Cd₃As₂, leading to a high ZT (0.5) of the p-type Cd₃As₂ being achieved at low temperature (500 K). Our work provides a new avenue towards high-performance thermoelectric materials based on topological Dirac semimetals.

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